

Sean Davis, MD, PhD

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Google Scholar | i10-index: 99 | h-index: 66 | 48656 citations (21253 since 2021; h-index 42 since 2021)

EDUCATION AND PROFESSIONAL EXPERIENCE

- 2021–Present **Professor of Medicine and Biomedical Informatics; Rifkin Bennis Chair of Cancer Informatics**
University of Colorado Anschutz School of Medicine
- 2017–2020 **Senior Associate Scientist**
Center for Cancer Research, National Cancer Institute
- 2009–2016 **Staff Scientist**
Center for Cancer Research, National Cancer Institute
- 2007–2008 **Research Fellow**
National Cancer Institute
- 2005–2007 **Research Fellow**
National Human Genome Research Institute
- 2002–2005 **Clinical Fellow**
Combined Johns Hopkins and National Cancer Institute Pediatric Hematology / Oncology Fellowship
- 1999–2002 **Pediatric Resident**
Children's Hospital and Regional Medical Center, University of Washington
- 1993–1999 **MD**
University of Pittsburgh School of Medicine
- 1995–1997 **PhD**
University of Pittsburgh Graduate School of Public Health, Department of Human Genetics
- 1989–1993 **B.S.E., With Honors**
Princeton University, Mechanical and Aerospace Engineering

LEADERSHIP AND SERVICE

Institutional Strategy

- 2026–Present Member, Clinical Predictive Model Governance Committee, UCHHealth (recurring Epic clinical-AI model review)

- April 2026 Author, *Implementing AI in Academic Medicine* — 196-page open AI governance reference distributed to CU Anschutz Chancellor, CIO, and Dean (amc-ai-governance)
- 2026-2027 Member, Education and Research AI Working Groups, Anschutz Medical Campus
- 2021–Present Founding Member and Faculty, Department of Biomedical Informatics, University of Colorado School of Medicine (chaired by Casey Greene)
- 2021–Present Associate Director for Informatics and Data Science, University of Colorado Comprehensive Cancer Center
- 2022–Present Member, University of Colorado Health System Clinical Intelligence Committee
- 2021–2023 Acting Lead, Center for Health AI software engineering team, University of Colorado School of Medicine
- 2021 Interviewer for Medical Oncology Fellowship program, University of Colorado Department of Medicine
- 2021–2023 Chair, open rank faculty search committee, Center for Health AI, University of Colorado School of Medicine
- 2021–2024 Deputy Director, Center for Health Artificial Intelligence, University of Colorado Anschutz School of Medicine
- 2021–2023 Led development of Alpine, the Anschutz Medical Campus high-performance computing environment (700+ researcher users in three years), in partnership with CU Anschutz OIT, School of Medicine IT, and CU Boulder Research Computing
- 2021–2024 Member, Compass Clinical Data Warehouse Steering Committee, University of Colorado School of Medicine
- 2021–Present Member, Governance Committee, University of Colorado Cancer Center
- 2021–Present Member, Executive Committee, University of Colorado Cancer Center
- 2015–2020 CCR Representative to CBIIT Strategic Planning Committee
- 2010–2020 Steering Committee, NCI Center of Excellence in Integrative Cancer Biology and Genomics
- 2010–2012 Scientific Liaison, Center for Cancer Research Bioinformatics Core
- 2009–2010 Chair, Center for Cancer Research Bioinformatics Planning and Implementation Committee
- 2009–2016 Sequencing Facility Steering and Review Committee, Center for Cancer Research, NCI

National & International Policy

- July 2026 Chair, NIH Center for Scientific Review study section panel
- 2026-Present Member, multiple NIH Common Fund Data Ecosystem (CFDE) governance and working groups — Steering Committee, SPARC-CFDE Quarterly, Admin Unified Auth Strategy WG, Ontology WG, Knowledge Graph WG, Marker Paper/Publication WG
- March 2026 Ad hoc reviewer, NIH Transformative R01 Program — 4 applications reviewed

- February 2026–Present Member, NCCN Digital Oncology Forum
- April 2026 Project Lead, NCI-DOE Genesis Mission (MOSSAIC): Advanced Computing and AI Foundations for Childhood and Young Adult Cancer Research
- June 2021 NIH/CSR special review of DSI-Africa U54 Center Grants
- 2020–Present Member of NHGRI AnVIL project Executive Committee
- 2016–2020 Founding Member, NIH Data Science Special Interest Group (>600 members)
- 2012–2019 Founding Member, NIH High Performance and Scientific Computing Working Group, responsible for expansion and maintenance of Biowulf, NIH's institutional high-performance computing environment (top-100 supercomputer worldwide)
- 2017 NIH Intramural Representative, NIH Data Commons working group
- December 2017 Co-organizer, NIH Hour of Code, Data Science Special Interest Group, NIH, Bethesda, MD
- November 2017 NIH Representative to US Department of Agriculture, Blueprint for USDA Efforts in Agricultural Animal Genomics, Beltsville, MD
- August 2017 NIH Intramural Representative, NIH Data Commons Review Committee
- February 2017 Organizer, NIH/NIST Medical Devices Cybersecurity Workshop, Bethesda, MD
- 2017–2020 Cancer Moonshot Blue Ribbon Panel Implementation Working Group, National Cancer Data Ecosystem
- January 2017 Organizer, Globus Data Platform Hackathon and Workshop, NIH, Bethesda, MD
- January 2017 NCI Representative, NHLBI TopMed Data Commons Planning Workshop
- 2016–2017 NCI Representative, NIH Data Commons Reference Dataset Working Group
- 2016 Presidential Subcommittee on AI and Machine Learning, Cancer Moonshot Initiative
- December 2015 NCI representative and panel member, FDA Informatics and Precision Medicine Workshop
- 2015–2017 NCI Cancer Cloud Pilot, Lead for Intramural Research Program evaluation and implementation
- 2014–2018 NIH and NCI Genomic Data Sharing Policy Implementation working groups
- 2014 NCI Center for Cancer Genomics Genomic Data Commons (GDC) Review Committee
- 2012–2016 High Throughput Molecular Data Working Group, National Cancer Institute

Community Infrastructure & Governance

- July 2026 Led migration of Bioconductor.org web infrastructure to Cloudflare R2 and Workers, reducing projected hosting costs from ~\$3,500–5,000/month to ~\$100/month (~\$45,000/year savings)
- 2025–2026 Organizing Committee, Bioconductor Annual Conference (BioC2026)
- April 2026 Co-Organizer, Bioconductor Spatial Data and Image Analysis Hackathon, Venice, Italy

- January 2026 Co-Organizer, NVIDIA × Carnegie Mellon University Biomedical AI Hackathon, Pittsburgh, PA
- 2008–2021 Organizer/Co-Organizer, Bioconductor Annual Meeting and Developer Conference (annual; Fred Hutchinson, Boston, Stanford, Seattle, Toronto, NYU, virtual)
- 2008–2020 Bioconductor Technical Advisory Board — shared responsibility for ongoing technical leadership of the Bioconductor Project
- May 2019 Co-organizer, Artificial Intelligence for Data Reuse, an NSF-funded symposium, Carnegie Mellon University, Pittsburgh, PA
- October 2018 Organizer, Informatics Training and Education for Cancer Centers, New Orleans, LA
- May 2018 Co-organizer, Renal Cell Cancer Hackathon, in collaboration with sv.ai and Google, San Francisco, CA
- July 2016 Co-organizer, Frontiers of Predictive Oncology and Computing Symposium, Washington, DC
- January 2016 Co-organizer, NCBI Genomics and Bioinformatics Hackathon
- January 2014 Organizer and Instructor, Bioinformatics Summer Course, Ribeirão Preto Medical School, University of São Paulo, Brazil
- April 2008 Organizer, European Bioconductor Developer Conference, Lausanne, Switzerland

TEACHING AND TRAINING

Recurring Courses

- 2024–Present **Frontiers of AI in Medicine** · *Course Director*
University of Colorado School of Medicine (medical student course)
- 2013–Present **Statistical Methods for Functional Genomics** · *Senior Instructor and Course Organizer*
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY
Annual two-week intensive; topics include exome sequencing, methylation arrays, CGH, public data access, and data integration
- 2005–2012 **AACR Molecular Biology in Clinical Oncology** · *Instructor*
Aspen & Snowmass, CO (annual, except 2008)
- 2015 **Harvard School of Engineering and Applied Science: CS290 Extreme Computing** · *Course Organizer and Faculty*
Harvard University, Cambridge, MA
- 2014 **Bioinformatics Summer Course** · *Organizer and Instructor*
Ribeirão Preto Medical School, University of São Paulo, Brazil
- Spring 2010 **Microarray Data Analysis Using R and Bioconductor** · *Instructor*
Department of Biostatistics, Bioinformatics, and Biomathematics, Georgetown University

Mentorship and Advising

- 2021–Present **Faculty Mentor, Medical Student Research Track**
University of Colorado School of Medicine
- 2021–Present **Faculty Member, Computational Biosciences Graduate Training Program**
University of Colorado School of Medicine
- 2015–2018 **PhD Thesis Committee** · *Vincent Laufer*
University of Alabama (graduated 2018)
- Summer 2016 **Undergraduate Research Mentor** · *Rosa Choe, computer science*
UC Berkeley (graduated 2019)
- Summer 2015 **Undergraduate Research Mentor** · *Olivia Zhang, computer science*
Princeton University (graduated 2020)
- Summer 2013 **Undergraduate Research Mentor** · *Peter Hansen, biology*
Cornell University (graduated 2018)

Selected Invited Presentations

- September 2026 **AI Agents for Epigenetics Research**
Agentic AI Workshop, Penn Epigenetics Institute, University of Pennsylvania, Philadelphia, PA
- July 2026 **AI Agents for Biomedical Knowledge Work**
University of Maryland Institute for Health Computing, North Bethesda, MD
- July 2026 **Agentic AI for Biomedical Research and Bioinformatics**
Pre-JSM workshop, Harvard University, Cambridge, MA
Three-hour invited session for ~25 attendees, including Bioconductor project leadership (Rafael Irizarry, Robert Gentleman, Vince Carey, Jean Yang, Lori Shepherd, Stephanie Hicks)
- July 2026 **AI-in-Software-Development Panel** · *Organizer and Panelist*
ITCR 2026 Annual Meeting, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY
- July 2026 **Two invited special lectures: microbiome data science and agentic AI for biomedical research**
Big Data Training for Cancer Research (R25 CA233429), Newport Beach, CA
- August 2025 **Bioconductor: Cancer Genomics — Integrative and Scalable Solutions**
Bioconductor Annual Conference (BioC2025) / NCI ITCR session
- November 2023 **Artificial Intelligence & Machine Learning**
Monroe Hospital, Monroe, Colorado
- September 2022 **Artificial Intelligence & Machine Learning**
Department of Medicine Grand Rounds, University of Colorado Anschutz Medical Campus
- December 2021 **Orchestra: a Cloud Platform and Ecosystem for Scalable Computational Training**
VIB/ELIXIR Framework workshop on cloud migration, Ghent, Belgium

- November 2021 **Bioinformatics, HPC and AI: Convergence, Perspectives and the Future for Biomedical Applications**
Expert panel, Supercomputer 2021 (SC21), St. Louis, MO
- September 2021 **Translation of AI and ML to a Learning Healthcare System**
Data Management Track, Bio-IT World, Boston, MA
- April 2021 **Leveraging Public Genomic Data Resources to Improve Science**
Virtual hands-on workshop for 100+ researchers, Human Heredity and Health in Africa (H3Africa) and H3Bionet
- September 2020 **sars2pack: An open software package providing fit-for-use COVID-19 data resources**
Cornell Weill School of Medicine, New York, NY
- July 2018 **Cloud computing approaches to genomic data science**
American Statistical Association, Joint Statistical Meeting, Vancouver, Canada
- January 2018 **A Data Ecosystem for Biomedical Big Data**
Grand Rounds, Wake Forest School of Medicine, Winston-Salem, NC
- January 2017 **A cloud-based data ecosystem for cancer research**
Dana Farber Cancer Institute, Boston, MA
- October 2016 **Democratizing access to Big Cancer Data**
Midatlantic Bioinformatics Conference, University of Pennsylvania, Philadelphia, PA
- November 2011 **High-resolution Views of the Cancer Genome Using Next-Generation Sequencing Approaches**
Lombardi Cancer Center, Georgetown University
- March 2012 **Featured Speaker, Bioinformatics for Medical Genetics Symposium**
American College of Medical Genetics, Charlotte, NC
- February 2012 **Advanced R and Bioconductor Workshop on High-Throughput Genetic Analysis**
Fred Hutchinson Cancer Research Center, Seattle, WA
- February 2009 **Genomics for the Pediatrician: An Overview of Genomics Technologies**
Pediatric Grand Rounds, Oklahoma University Health Sciences Center, Oklahoma City, OK

PROGRAM OF RESEARCH — SELECTED PUBLICATIONS

Full publication list: Google Scholar (h-index 66, 48656 citations; h-index 42 and 21253 citations since 2021). Software tools downloaded by 50,000+ users/year. Complete bibliography at end of document.

Data Infrastructure & Bioconductor

Orchestrating high-throughput genomic analysis with Bioconductor

Nature Methods, 2015 · 4313 citations — *Landmark reference describing the Bioconductor project infrastructure*

Bioconductor's Computational Ecosystem for Genomic Data Science in Cancer

Methods in Molecular Biology, 2025

MultiAssayExperiment: Software for the Integration of Multi-Omics Experiments in Bioconductor

Cancer Research, 2017 · 148 citations

GenomicDataCommons: a Bioconductor Interface to the NCI Genomic Data Commons

Bioinformatics, 2017 · 11 citations

GenomicSuperSignature facilitates interpretation of RNA-seq experiments through robust, efficient comparison to public databases

Nature Communications, 2022 · 14 citations

BugSigDB captures patterns of differential abundance across a broad range of host-associated microbial signatures

Nature Biotechnology, 2024 · 23 citations

GEOquery: a bridge between the Gene Expression Omnibus (GEO) and BioConductor

Bioinformatics, 2007 · 3505 citations — *Canonical R/Bioconductor interface to NCBI GEO; widely-used community resource*

RCircos: an R package for Circos 2D track plots

BMC Bioinformatics, 2013 · 1013 citations

Translational Research: Cancer Genomics & Oncology

Pan-cancer genome and transcriptome analyses of 1,699 paediatric leukaemias and solid tumours

Nature, 2018 · 1016 citations — *International pediatric cancer consortium*

Super-enhancers delineate disease-associated regulatory nodes in T cells

Nature, 2015 · 424 citations

Meta-analysis of 22,710 human microbiome metagenomes defines an oral-to-gut microbial enrichment score and associations with host health and disease

Nature Communications, 2025 · 23 citations

Integration of 168,000 samples reveals global patterns of the human gut microbiome

Cell, 2025 · 133 citations

A pivotal role for Wnt antagonists in constraining Wnt activity to promote digit joint specification

bioRxiv (preprint), 2025

ASXL3 Is a Novel Pluripotency Factor in Human Respiratory Epithelial Cells and a Potential Therapeutic Target in Small Cell Lung Cancer

Cancer Research, 2017 · 35 citations

Lineage of origin in rhabdomyosarcoma informs pharmacological response

Genes & Development, 2014 · 130 citations

Policy, Governance & Data Ethics

The Common Fund Data Ecosystem (CFDE)

bioRxiv (preprint), 2026 · 1 citations — *Senior and corresponding author; descriptor paper for the NIH Common Fund Data Ecosystem*

Lifecycle-aware evaluation of biomedical data ecosystems

Zenodo (preprint; submitted to Cell Trends Open), 2026 — *Multidimensional evaluation framework for NIH CFDE; preprint at doi.org/10.5281/zenodo.20314128*

From Patient Engagement to Precision Oncology: Leveraging Informatics to Advance Cancer Care

Yearbook of Medical Informatics, 2020 · 20 citations

Resources for Interpreting Variants in Precision Genomic Oncology Applications

Frontiers in Oncology, 2017 · 24 citations

Public data and open source tools for multi-assay genomic investigation of disease

Briefings in Bioinformatics, 2016 · 65 citations

Ten simple rules for large-scale data processing

PLOS Computational Biology, 2022 · 5 citations

Community Building & Training

Learning and teaching biological data science in the Bioconductor community

PLOS Computational Biology, 2025 · 7 citations

Orchestrating a community-developed computational workshop and accompanying training materials

F1000Research, 2018

Exploring Cancer in Colorado using a novel data platform: the ECCO experience

medRxiv, 2026 — *Community data platform for cancer disparities research*

SELECTED SOFTWARE

Software tools downloaded by **50,000+ users per year**.

cfde-atlas

Production chatbot over the NIH Common Fund Data Ecosystem (50+ C2M2 tables); demoed to NIH leadership.

cu-research-intelligence

Research-intelligence platform mining OpenAlex + NIH RePORTER; two organic institutional customers (CU Cancer Center CCSG, Rady Center).

orchestraplatform

Cloud platform and ecosystem for scalable computational training; sole steward / maintainer / funder.

biomedical-agent-kg

Knowledge-graph construction toolkit for biomedical agentic AI applications.

scriptorium

Agentic operating system for scholarly writing — citation audit, reviewer simulation, structural analysis.

quartobot

Quarto-based scholarly publishing toolchain; manubot successor, on PyPI.

fda-approval-app

FDA approval lookup for clinical-trials operations (openFDA + LLM arbiter).

pubmed-grader

Bibliometric dashboard for cancer center portfolio analysis (iCite + Unpaywall + Altmetric).

nextflow_telemetry

Cloud-native observability stack for Nextflow workflow runs.

lifeos-template

Public template for AI-augmented research operating systems.

awesome-single-cell

Community-curated reference list of single-cell software and data resources (3,700+ GitHub stars).

awesome-cancer-variant-resources

Community-maintained catalog of cancer-variant clinical knowledgebases and databases.

OmicIDX

Cloud-native metadata index for 80M+ SRA runs and 8M GEO samples.

GEOquery

Canonical R/Bioconductor interface to NCBI Gene Expression Omnibus.

Genomic Data Commons

R/Bioconductor client for the NCI Genomic Data Commons.

BiocPkgTools

Tools for analyzing and reporting on the Bioconductor package ecosystem.

GenomicSuperSignatures

Interpret RNA-seq experiments via robust comparison to public databases.

SRADB

SQLite-backed query interface to the NCBI Sequence Read Archive.

GEOmetadb

SQLite-backed query interface to NCBI GEO sample metadata.

biomaRt

Programmatic R interface to BioMart databases (e.g., Ensembl).

RCircos

Circular genome-data visualization in R (CRAN).

methyumi

R/Bioconductor infrastructure for Illumina methylation array analysis.

ACME

Algorithms for calculating microarray enrichment (R/Bioconductor).

ngCGH

Next-gen sequencing copy-number analysis (early HTS-era tool).

sars2pack

Fit-for-use COVID-19 data resources for R (2020 pandemic response).

state-cancer-profiles

Programmatic access to NCI State Cancer Profiles data.

AWARDS AND HONORS

- 2018 NIH Director's Award
- 2017 NCI Technology Transfer Award
- 2016 NCI Technology Transfer Award
- 2016 U.S. Department of Health and Human Services Director's Award
- 2015 NIH Director's Award
- 2012 Staff Scientist/Staff Clinician Travel Award, Center for Cancer Research, NCI
- 2002–2007 NIH General Loan Repayment Program
- 2002 Family-Centered Care Award, University of Washington, Children's Hospital of Seattle
- 1995 W.M. Keck Fellowship for Advanced Scientific Computing
- 1989 National Merit Scholar
- 1988 Pennsylvania Governor's School for Science

FUNDING

Current

Comprehensive genetic ancestry annotation of existing human molecular data

R01 Supplement | NIH | 12/01/2025–11/30/2030 | Total: \$214K

PI: Alex Krasnitz Role: *Co-Investigator*

Site: Cold Spring Harbor Laboratory

Annotates public human molecular data with ancestry predictions to improve understanding of diversity across human genomics datasets.

Big Data Training for Cancer Research

R25 CA233429 | NIH | 09/01/2024–08/31/2029 | Total: \$1615K

PI: Min Zhang Role: *Co-Investigator*

Site: University of Colorado Denver

Supports a two-week intensive, hands-on cancer informatics training workshop for biologists learning large-scale cancer data analysis.

Exploiting Public Metagenomic Data to Uncover Cancer-Microbiome Relationships

R01 CA230551 | NCI | 09/01/2024–08/31/2029 | Total: \$2219K

PI: Levi Waldron Role: *Co-Investigator*

Site: University of Colorado Denver

Investigates the role of the human microbiome in cancer using published metagenomic data and returns improved, more reusable resources to the research community.

Integration, Dissemination, and Evaluation (BRIDGE) Center for the NIH Bridge to Artificial Intelligence (BRIDGE2AI) Program

U54 HG012513 | NIH | 07/01/2022–04/30/2026 | Total: \$10518K

PI: Monica Cecilia Munoz-Torres Role: *Co-Investigator*

Site: University of Colorado Denver

Coordinates the BRIDGE2AI program to ensure data are ethically sourced, machine-understandable, and interoperable for AI and machine learning applications.

Cancer Genomics: Integrative and Scalable Solutions in R/Bioconductor

U24 CA289073 | NIH | 07/01/2024–06/30/2029 | Total: \$5225K

PI: Levi Waldron, Sean Davis Role: *MPI*

Site: University of Colorado Denver

Develops, expands, and sustains Bioconductor software and data resources for cancer genomics research, analysis, and training.

Oculomics as a Biomarker for Comprehensive and Non-Invasive Patient Health Assessment

AWD-AAI2401 | Anschutz Acceleration Initiative, Anschutz Foundation | 07/01/2024–02/28/2029 | Total: \$6979K

PI: Sean Davis Role: *Co-Investigator*

Site: University of Colorado Anschutz School of Medicine

Uses ophthalmic imaging and AI to identify non-invasive biomarkers for systemic disease risk and earlier patient assessment.

CONNECT: Collaborative Network for Nurturing Ecosystems of Common Fund Team Science

U54 OD036472 | NIH | 04/01/2024–03/31/2029 | Total: \$2937K

PI: Jake Yue Chen, Sean Davis, Casey S. Greene, Peipei Ping, Wei Wang Role: *MPI*

Site: University of Colorado Denver

Builds collaborative infrastructure and coordination capacity for the NIH Common Fund Data Ecosystem so investigators can better reuse and integrate Common Fund resources.

Leveraging whole-exome sequence data from diverse biobanks and cohorts to study rare coding variation in prostate cancer

R01 CA276826 | NCI | 09/01/2023–08/31/2028 | Total: \$450K

PI: Christopher Alan Haiman Role: *Co-Investigator*

Site: University of Colorado Denver

Studies rare and common genetic variation associated with prostate cancer susceptibility, progression, and staging using globally diverse biobank cohorts.

University of Colorado Cancer Center Support Grant

P30 CA046934 | NCI | 02/01/2023–01/31/2027 | Total: \$4137K

PI: Richard D. Schulick Role: *Co-Investigator*

Site: University of Colorado Denver

Supports the University of Colorado Cancer Center's cancer discovery, translation, patient-centered care, training, and catchment-area impact.

Pending

Genetic Ancestry knowledgebase for human molecular data in SRA

NIH pending | NIH | 12/01/2026–11/30/2030 | Total: \$144K

PI: Alex Krasnitz Role: *Co-Investigator*

Site: Cold Spring Harbor Laboratory

Proposes a repository of ancestry annotations for human Sequence Read Archive data to enable ancestry-aware analyses at population scale.

A comprehensive biomedical metadata knowledgebase to democratize data discovery

R24 HG014772 | NHGRI | 12/01/2025–11/30/2029 | Total: \$2142K

PI: Arjun Krishnan Role: *Co-Investigator*

Site: University of Colorado Denver

Proposes a metadata knowledgebase to support metadata enrichment, deep search, and machine learning over public biomedical datasets.

Foundational Agentic Workflows in Bioconductor

OS4LS Track 2 (Sloan/Schmidt Sciences) | Open Source for Life Sciences (OS4LS) | 01/01/2027–12/31/2028

PI: Levi Waldron Role: *Co-Investigator*

Site: University of Colorado Denver

Proposes agentic-AI-assisted tooling to sustain and extend foundational Bioconductor libraries and ecosystem infrastructure. Collaborative US (CU, CUNY) / Italy proposal, submitted 2026-07-21.

A Knowledgebase Streamlining Access to Public Omics Data through Curated Metadata

R24 (pending) | NIH | 12/01/2025–11/30/2029 | Total: \$70K

PI: Arjun Krishnan Role: *Co-Investigator*

Site: University of Colorado Denver

Proposes a public knowledgebase that uses controlled vocabularies and ontologies to improve access to and reuse of public omics data.

Completed

Bioconductor: high quality training and support for a worldwide community

Chan Zuckerberg Initiative, Essential Open Source Software for Science, Cycle 4 | 01/01/2022–12/31/2023

PI: Aedin Culhane Role: *Co-Investigator*

Uses technology to extend Bioconductor data science and bioinformatics training efforts to better support diversity, inclusivity, and global reach.

BigRNA: Accelerating Research By Increasing Access and Use of Public Transcriptomic Data

Amazon Research Grant | 06/01/2019–06/01/2020

PI: Sean Davis Role: *PI*

The BigRNA project reprocesses and distributes publicly available transcriptomic data to facilitate data reuse, reanalysis, and machine learning applications.

Symposium on Personal Control of Genomic Data for Research

Cancer Moonshot Initiative | September 2019

PI: Sean Davis, on behalf of the Network for Direct Patient Engagement Role: *PI*

Brought together patients, advocates, lawyers, ethicists, medical care providers, and data scientists to share expertise on personal control of genomic data.

Increasing the value of public transcriptomic data through harmonized processing and metadata

XSEDE (PSC) MCB200027 | 01/30/2020–01/29/2021

PI: Sean Davis Role: *PI*

Allocation to prototype using XSEDE resources to perform large-scale, harmonized omics data processing, including transcriptomics and metagenomics.

Increasing the value of public transcriptomic data through harmonized processing and metadata

Administrative Supplement to U01 CA230551 | 10/01/2021–09/30/2022

PI: Levi Waldron Role: *Co-Investigator*

Adds support for >100,000 homogeneously processed microbiome samples to the public microbiome data collection.

EDITORIAL RESPONSIBILITIES

2015–Present Editor, f1000research Bioconductor Channel

2010–Present Associate Editor, BMC Bioinformatics

2009 Book reviewer, CRC Press

Peer Reviewer — selected journals:

- Bioinformatics
- Cancer Research
- Genome Research
- Molecular Cancer Research
- Nucleic Acids Research
- PLOS One
- Gigascience
- BMC Bioinformatics
- Clinical Cancer Research
- Genomics
- Nature Methods
- PLOS Computational Biology
- Genetic Epidemiology
- Briefings in Bioinformatics

PUBLICATIONS

Complete list. Google Scholar link provides citation metrics and open-access links.

Davis, S. (2026). *Implementing AI in Academic Medicine: A guide to governance and principled practice*. <https://seandavi.github.io/amc-ai-governance/>

Lowery, J. T., Alquaddoomi, F., Rubinetti, V., Burus, T., Jardine, C., Warren, A. C., Walsh, J., Borrayo, E., & Davis, S. (2026, February 4). *Exploring Cancer in Colorado using a novel data platform: the ECCO experience*. <https://doi.org/10.64898/2026.02.03.26345489>

Lange, E. M., Ganesh, V. A., Ivich, A., Rubinetti, V., Greene, C., & Davis, S. (2026, May 20). *Lifecycle-aware evaluation of biomedical data ecosystems*. Zenodo. <https://doi.org/10.5281/zenodo.20314128>

Crowell, H., Marconato, L., Billato, I., Calgaro, M., Castelo, R., Ceccaroni, R., Chen, C., Danaher, P., Davis, S., Emons, M., Grajeda, G., Gruson, H., Gunz, S., Henao, J., Irizarry, R., Kharuk, S., Lazic, D., Manukyan,

- A., Moles-Seró, P., ... Risso, D. (2026, June 14). *Bioconductor Spatial Data and Image Analysis Hackathon*. BioHackrXiv (Center for Open Science). https://doi.org/10.37044/osf.io/9ej32_v1
- Abdill, R. J., Graham, S. P., Rubinetti, V., Ahmadian, M., Hicks, P., Chetty, A., McDonald, D., Ferretti, P., Gibbons, E., Rossi, M., Krishnan, A., Albert, F. W., Greene, C. S., Davis, S., & Blekman, R. (2025). Integration of 168,000 samples reveals global patterns of the human gut microbiome. *Cell*, 188(4), 1100–1118.e17. <https://doi.org/10.1016/j.cell.2024.12.017>
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- Huang, Y. P., Harmon, L., Deering-Gardner, E., Ma, X., Harsh, J., Xue, Z., Wen, H., Ramos, M., Davis, S., & Triche, T. J. J. (2025). bamSliceR: a Bioconductor package for rapid, cross-cohort variant and allelic bias analysis. *Bioinformatics Advances*, 5(1), vbaf98. <https://doi.org/10.1093/bioadv/vbaf098>
- Manghi, P., Antonello, G., Schiffer, L., Golzato, D., Wokaty, A., Beghini, F., Mirzayi, C., Long, K., Gravel-Pucillo, K., Piccinno, G., Gamboa-Tuz, S. D., Bonetti, A., D'Amato, G., Azhar, R., Eckenrode, K., Zohra, F., Giunchiglia, V., Keller, M., Pedrotti, A., ... Waldron, L. (2025). Meta-analysis of 22,710 human microbiome metagenomes defines an oral-to-gut microbial enrichment score and associations with host health and disease. *Nature Communications*, 17(1), 196. <https://doi.org/10.1038/s41467-025-66888-1>
- Ramos, M., Shepherd, L., Sheffield, N. C., Mahmoud, A., Pages, H., Wokaty, A., Righelli, D., Risso, D., Davis, S., Oh, S., Waldron, L., Morgan, M., & Carey, V. (2025). Bioconductor's Computational Ecosystem for Genomic Data Science in Cancer. *Methods in Molecular Biology (Clifton, N.J.)*, 2932, 1–46. https://doi.org/10.1007/978-1-0716-4566-6_1
- Huang, B.-L., Davis, S., Koyama, E., Pacifici, M., & Mackem, S. (2025, July 21). *A pivotal role for Wnt antagonists in constraining Wnt activity to promote digit joint specification*. <https://doi.org/10.1101/2025.07.17.665381>
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